Band-1 Hadamard Order 4196 — Miyamoto (Double Paley-I seed H_1048); Gap 2052 Still OPEN

H_524/H_1048/C_1050/H_4196 PASS; 4196 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4132_Miyamoto`; `2026-09-15_Band1_Hadamard_4036_Miyamoto` (Double Paley-I seed pattern).
Lab. `/workspace/experiments/band1/hadamard_4196_miyamoto/`.
Cutoff. 2026-09-15T14:45:48Z (~10:45AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 524, 1048 and 4196 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1048} (H_{524} Paley-I $q = 523$, then Sylvester double). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4196 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4195 = 5 \cdot 839$ not prime); Paley-II **FAIL** ($q = 2097 = 3^2 \cdot 233$ not prime); Kronecker **FAIL** ($4196 = 4 \cdot 1049$, 1049 not Hadamard-admissible). Seed triage for H_{1048}: Paley-II **FAIL** ($q = 523 \equiv 3 \pmod 4$, needs $\equiv 1$); Paley-I direct **SKIP** ($q = 1047 = 3 \cdot 349$ not prime); Double(Paley-I) **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4132 on box. This increment ships **order 4196** ($4196 = 4 \cdot 1049$, 1049 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1048} (H_{524} from Paley-I $q = 523$, Sylvester-doubled). Symmetric Paley conference C_{1050} over prime 1049; $H_{4196} = \text{Miyamoto}(H_{1048}, C_{1050})$. Unit tests **34/34 PASS**. SPK next gap remains **2052** (OPEN). Order 4196 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4196; Miyamoto; Paley type I; Sylvester double; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{524} constructed and verified (Paley type I, $q = 523$). 2. H_{1048} constructed and verified (Sylvester double of H_{524}). Paley-II path for H_{1048} **rejected** ($q = 523 \equiv 3 \pmod 4$). 3. C_{1050} constructed and verified (Paley conference, $q = 1049$): $CC^\top = 1049I$, zero diagonal. 4. H_{4196} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $4196 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4196 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4196 = 4 \cdot 1049$
$q = 1049$	prime, $1049 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1048}
Needed conference	order $1050 = q + 1$
H_{1048} path (used)	Double(Paley-I): H_{524} Paley-I $q = 523$, Sylvester double $\rightarrow H_{1048}$
H_{1048} Paley-II	**FAIL**: $2(q+1) = 1048 \Rightarrow q = 523 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1048} Paley-I direct	**SKIP**: $q + 1 = 1048 \Rightarrow q = 1047 = 3 \cdot 349$ not prime
Conference path	Symmetric Paley over prime 1049
Paley-I for 4196	**REJECTED**: $q' = 4195 = 5 \cdot 839$ not prime
Paley-II for 4196	**REJECTED**: $q' = 2097 = 3^2 \cdot 233$ not prime
Kronecker for 4196	**REJECTED**: only 4×1049; $1049 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{524}	Paley type I, $q = 523$	0	0.2004	**PASS**
H_{1048}	Sylvester double of H_{524}	0	0.5218	**PASS**
C_{1050}	Paley conference, $q = 1049$	0	1.0156	**PASS**
H_{4196}	Miyamoto(H_{1048}, C_{1050})	0	33.2099	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 36.1227 s. Compressed ± 1 matrices saved as `H524_pm1.npz`, `H1048_pm1.npz`, `C1050_pm1.npz`, `H4196_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020** Paley-II + **1032** Paley-I; **exclude** 4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{524} (Paley-I) $\cup$ verified H_{1048} (Sylvester-double) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4196 remains a Miyamoto-only bonus.

Quantity	Value
Known count	964
4196 in SPK	**False** (Miyamoto-only)
1048 in SPK	**True** (Sylvester-double of Paley-I)
524 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**Yes** (like 4132/4084/3716/3524/…)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

34/34 PASS (Sylvester/Paley/Miyamoto controls; $q = 523/1049$ arithmetic; Double Paley-I seed for H_{1048}; Paley-II seed rejection; SPK path rejection for 4196; Miyamoto-only arith guard; onbox seeds include 1032/1020 / exclude 4132/4084/4196; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 4196 is a **Miyamoto-only bonus (NOT in SPK)**.

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1093 \to n = 4372$ PaleyI($q = 1091$); or $q = 1097 \to n = 4388$ Double(PaleyI half=548); or $q = 1117 \to n = 4468$ PaleyII($q = 557$)) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_4196_miyamoto/` · *Script:* `verify_hadamard_4196.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.